Lab 2 An Experimental Exploration: Matching and Compare Signals

Overview of the Lab: In this lab we are going to compare periodic signal. We are going to experiment and start thinking about how can we compare one signal to another. How can we say one signal is more similar to another signal? I will be upfront now, this lab is not going to fully give you those answers, but it will lay the foundation for comparing signals. Maybe one day, you have an unknown signal and you want to know how similar it is to another signal. You are probably asking yourself one of the following: "why would I want to do that?", "That sounds super boring!", "No one will ever need to do this!". Therefore, let us give you some contexts to an application of how this can be potentially used.

Contextually Example Let's pose you the following scenario. We have a database or set of templates that are associated to different heart diseases from electrocardiograms (Heart Signals). We can see these diseased heart signals through the visualization within Figure 1 below called "Different Heart Signals Indicating Disease". We have our beloved family member's heart signal. We are posed the following question: Which cardiovascular disease is inflicting our beloved family member? Well, maybe a matching algorithm can aid in determining what components exist within our beloved family's heart signal and how it compares to the disease signal.

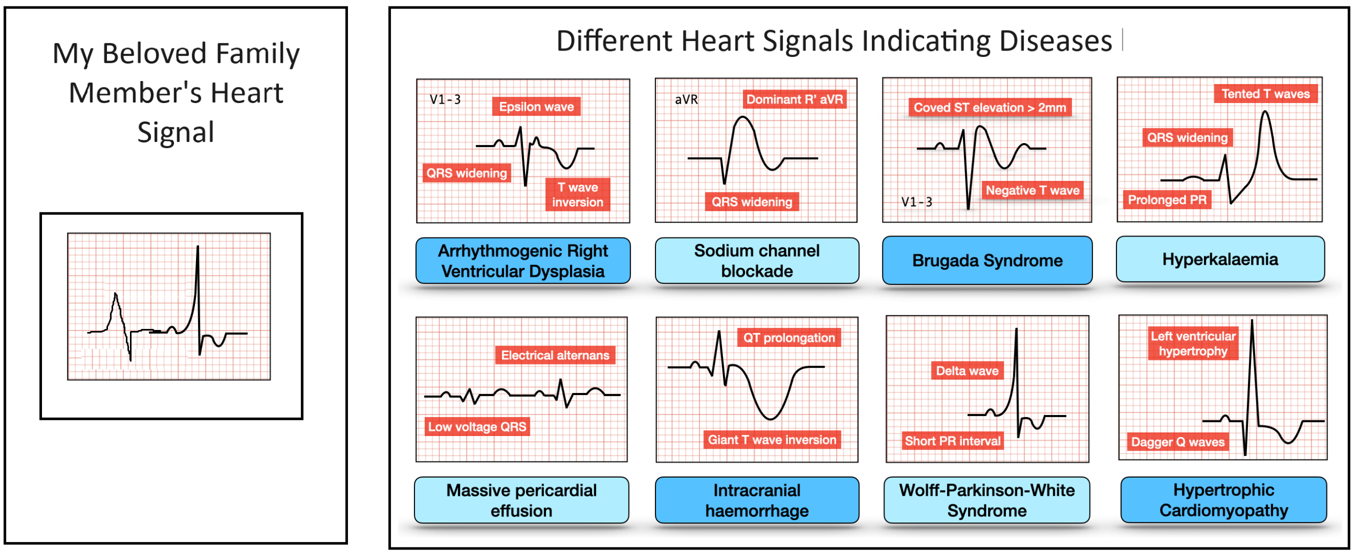

Figure 1: Contextual Example for the Need to Compare Signals

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1 Experimental Design

Now that we are motivated on developing a way to compare signals, let's begin a computational experiment by making our own database of signals and examine different algorithmic approaches for comparing signals. We will need to start at the most basic level first and nothing too complex. Let's set our goal to determine a single attribute from the signal. For this experiment we are going to try to compare signals with different frequencies. Therefore, we will hold phase and amplitude constant with a sampling rate of 1000Hz.

1.1 Step 1: Template Database

Let's make our own database with 14 different signals that start at 1 Hz and go 98.5 Hz which increments in 7.5 Hz. We will assume signals are composed of a cosine wave form with zero phase, amplitude equal to one, and each signal is 3 seconds long.

Answer the following Questions:

1. Report exactly what is being stored within in 'f' and the dimesions.
2. Report exactly what is being stored in 'Fs' and the dimensions.
3. Describe what is in "TemplateDataBase" and the dimensions.

MATLAB Template Database Code:

```

clear all
close all
clc
%-------------------------------------
% Comparing Waveforms With Real Numbers
%-------------------------------------
    f=??                                % Hint: this should be a vector or array
    fs=???;                             % Scalar Value
    dt=;                                % Scalar Value
    t=????                              % Time Vector
    A=1;

    for N=1:length(f)
        TemplateDataBase(N,:)=A*cos(2*pi*??*t);  % Creates Database where f is specifically indexed by N
    end
```

1.2 Step 2: Create Your Own Method to Compare with a Known Cosine Signal

Create a new "unknown" or "ground truth" signal (We technically know the signal. Since we created it, but are using this to validate the algorithm). This ground truth signal will allow us to compare a specific signal to each one of our database templates. Let's make it very easy to start, where our ground truth signal, SignalGT, has an amplitude of 1 and frequency equal to the second component within the array of 'f' (i.e. $f_{GT} = f(2)$).

MATLAB Code to Explore Our Own Signal Comparison:

```
    A=??
    f_GT=??;
    Signal_GT=A*cos(2*pi*f_GT*t);                         % Ground Truth Signal

    for N=1:length(f)                                         % Iterate through the known signal and your database
        SignalCompare(N)=   ???? SignalGT ??? TemplateDataBase(N,:) ???  % Store a single value metric that provides context on the strength of the match or comparision.
    end
```

<u>Answer the following Questions:</u>

1. What is the value of f_{GT}
2. What are the dimensions of $Signal_{GT}$
3. What was the method you developed to compare the two signals? Why did you use this method. Discuss your logic in detail (More than 4 sentences). The logic doesn't have to be correct but I am looking for your thought process. Report and provide plots.

1.3 Step 3: Comparing Energy Between the Signals

Let's think about relating the energy or power within the two signals as a function of time. When two things are similar they magnify and if two things are different they attenuate. Thus, the energy or power can magnify or attenuate. One way we can do this is by simply taking the dot product of the two signals, by

$$y(t) = X_{GT}[n] \cdot X_{TD}[n] \tag{1}$$

$$y(t) = X_{GT}[1]X_{TD}[1] + X_{GT}[2]X_{TD}[2] + X_{GT}[3]X_{TD}[3] + \ldots + X_{GT}[N]X_{TD}[N] \tag{2}$$

where $X_{GT}[n]$ is your ground truth signal and $X_{TD}[n]$ is one of signals from your template database. We can note that we can multipling element by element for the dot product operation. In order to get a single measurement of engergy, we can take the sum over the entire time period of the two signals that are 3 seconds long.

$$E(t) = \sum_{n=1}^{n=N} X_{GT}[n] \cdot X_{TD}[n] \tag{3}$$

$$E(t) = \sum_{n=1}^{n=N} X_{GT}[1]X_{TD}[1] + X_{GT}[2]X_{TD}[2] + X_{GT}[3]X_{TD}[3] + \ldots + X_{GT}[N]X_{TD}[N] \tag{4}$$

where we can get the total energy or power (if normalized over time).

MATLAB Code to Explore Our Own Signal Comparison:

```
f_gt=??? ;
SignalGT=A*cos(2*pi*f_gt*t);

    for N=1:length(f)
        SignalEnergy(N)=??? SignalGT ??? TemplateDataBase(N,:) ??;
    end

figure
stem(f,SignalEnergy)
grid on
title('Comparing Signals with Real Numbers')
```

Answer the following Questions:

1. Using $f_{GT} = f(2)$ as your first experimental frequency. Provides the plots and described what happened.
2. Alter the frequency to other values within the vector f. Provides the plots and described what happened.
3. Alter the frequency to other values near the elements within the vector f but not exactly the values within the array. Start with deviations of .1 Hz, .25 Hz, .5Hz, 1 Hz, 2Hz, 3 Hz, 5Hz. Provides the plots and described what happened in great detail! (5-8 Sentences. Analyze and think about what is happening)
4. Significantly decrease and increase your time vector (1 sec, 20 seconds). Provides the plots and described what happened.

1.4 Step 4: Comparing Energy With Real and Imaginary Components

Let's think about how using imaginary components can be utilized that relating the energy or power within the two signals as a function of time. Let us alter our ground truth signal function, where

$$X_{GT} = cos(2\pi f_{gt}t) + i \cdot sin(2\pi f_{gt}t) \tag{5}$$

Lets preform the as process,where we take the sum over the entire time period of the two signals where the time vector is 3 seconds long, by

$$E(t) = \sum_{n=1}^{n=N} X_{GT}[n] \cdot X_{TD}[n] \tag{6}$$

$$E(t) = \sum_{n=1}^{n=N} X_{GT}[1]X_{TD}[1] + X_{GT}[2]X_{TD}[2] + X_{GT}[3]X_{TD}[3] + \ldots + X_{GT}[N]X_{TD}[N], \tag{7}$$

where we can get the total energy or power (if normalized over time).

MATLAB Code to Explore Our Own Signal Comparison:

```
 f_gt=??? ;
SignalGT=????? + ????;

    for N=1:length(f)
        SignalEnergy(N)=??? SignalGT ??? TemplateDataBase(N,:) ??;
    end

figure
stem(f,SignalEnergy)
grid on
title('Comparing Signals using Complex Numbers')
```

Answer the following Questions:

1. Using $f_{GT} = f(2)$ as your first experimental frequency. Provides the plots and described what happened.
2. Alter the frequency to other values within the vector f. Provides the plots and described what happened.
3. Alter the frequency to other values near the elements within the vector f but not exactly the values within the array. Start with deviations of .1 Hz, .25 Hz, .5Hz, 1 Hz, 2Hz, 3 Hz, 5Hz. Provides the plots and described what happened in great detail! (5-8 Sentences. Analyze and think about what is happening)
4. Significantly decrease and increase your time vector (1 sec, 20 seconds). Provides the plots and described what happened.